To get around this problem, we will make two modifications to the sinc function in (b), resulting in the waveform shown in (c). First, it is truncated to $M+1$ points, symmetrically chosen around the main lobe, where M is an even number. All samples outside these $M+1$ points are set to zero, or simply ignored. Second, the entire sequence is shifted to the right so that it runs from 0 to M. This allows the filter kernel to be represented using only *positive* indexes. While many programming languages allow *negative* indexes, they are a nuisance to use. The sole effect of this $M/2$ shift in the filter kernel is to shift the output signal by the same amount.

Since the modified filter kernel is only an approximation to the ideal filter kernel, it will not have an ideal frequency response. To find the frequency response that is obtained, the Fourier transform can be taken of the signal in (c), resulting in the curve in (d). It's a mess! There is excessive ripple in the passband and poor attenuation in the stopband (recall the Gibbs effect discussed in Chapter 11). These problems result from the abrupt discontinuity at the ends of the truncated sinc function. Increasing the length of the filter kernel does not reduce these problems; the discontinuity is significant no matter how long M is made.

Fortunately, there is a simple method of improving this situation. Figure (e) shows a smoothly tapered curve called a **Blackman window**. Multiplying the truncated-sinc, (c), by the Blackman window, (e), results in the **windowed-sinc** filter kernel shown in (f). The idea is to reduce the abruptness of the truncated ends and thereby improve the frequency response. Figure (g) shows this improvement. The passband is now flat, and the stopband attenuation is so good it cannot be seen in this graph.

Several different windows are available, most of them named after their original developers in the 1950s. Only two are worth using, the **Hamming window** and the **Blackman window** These are given by:

EQUATION 16-1
The Hamming window. These windows run from $i = 0$ to M, for a total of $M+1$ points.

$$w[i] = 0.54 - 0.46\cos(2\pi i/M)$$

EQUATION 16-2
The Blackman window.

$$w[i] = 0.42 - 0.5\cos(2\pi i/M) + 0.08\cos(4\pi i/M)$$

Figure 16-2a shows the shape of these two windows for $M = 50$ (i.e., 51 total points in the curves). Which of these two windows should you use? It's a trade-off between parameters. As shown in Fig. 16-2b, the Hamming window has about a 20% faster *roll-off* than the Blackman. However,

FIGURE 16-1 (facing page)
Derivation of the windowed-sinc filter kernel. The frequency response of the ideal low-pass filter is shown in (a), with the corresponding filter kernel in (b), a sinc function. Since the sinc is infinitely long, it must be truncated to be used in a computer, as shown in (c). However, this truncation results in undesirable changes in the frequency response, (d). The solution is to multiply the truncated-sinc with a smooth window, (e), resulting in the windowed-sinc filter kernel, (f). The frequency response of the windowed-sinc, (g), is smooth and well behaved. These figures are not to scale.